

Introduction to multi(transcript+gen)omics

Integrating Genomics and
Transcriptomics

Course Curriculum

- 1. Introduction to Multi-Omics (Genomics + Transcriptomics)
- 2. Genetic Variation Regulation - Quantitative Trait Loci
- 3. Gene Co-Expression Modules (WGCNA)
- 4. Multi-Omics Factor Analysis (MOFA2)
- 5. Introduction to Machine Learning for Biology
- 6. Supervised Learning and Predictive Modeling
- 7. Model Interpretability and Explainable AI
- 8. Subgroup Discovery

<https://github.com/vladimirkovacevic/multiomics>

Today's Lesson Overview

- Linear Algebra Foundations
 - Vectors, matrices, decompositions - the math behind the methods
- Multiomics data preparation workflow
- Understanding Multi-Omics Data
 - Genomics and transcriptomics - what we measure and why
 - Our course dataset: 300 samples, 100 genes, 50 SNPs, ASD vs.control
 - Data Preprocessing
 - Loading, validation, and QC checks
- Key Theoretical Concepts
 - Why integration matters and what comes next

LINEAR ALGEBRA FOUNDATIONS FOR MULTI-OMICS

1. Vectors and Matrices

- Vectors
 - Representation of gene expression profiles, feature vectors, embeddings
 - Example: One patient's expression across 100 genes = 100-dimensional vector
 - Operations: vector addition, scalar multiplication, dot product
- Matrices
 - Expression matrices: genes (rows) $\times$ samples (columns)
 - Example: 100 genes $\times$ 300 samples = 100 $\times$ 300 matrix
 - Operations: addition, scalar multiplication, matrix multiplication
 - Shape and dimensionality - critical for MOFA2, PCA

2. Matrix Operations

- Transpose (A^T)
 - Flip rows and columns: $(100 \text{ genes} \times 300 \text{ samples})^T = (300 \times 100)$
 - Used in inner products and covariance calculations
- Matrix Multiplication ($A \cdot B$)
 - Combining features and projecting data
 - Rule: $(m \times n) \cdot (n \times p) = (m \times p)$ - inner dimensions must match!
 - - `np.multiply()` = element-wise (Hadamard product)
 - - `np.dot()` or `@` = matrix multiplication
- Element-wise vs. Matrix Operations
 - Element-wise: multiply with scalar, normalize each gene independently
 - Matrix: transform entire dataset together
- Centering: Shifts data so the mean is zero - subtract mean from all values

3. Special Matrices

- Identity Matrix (I)
 - 1s on diagonal, 0s elsewhere
 - $A \cdot I = I \cdot A = A$ (like multiplying by 1)
 - Used in regularization: $(X^T X + \lambda I)$
- Diagonal Matrices
 - Only diagonal elements are non-zero
 - Feature scaling: $D \cdot X$ scales each row of X
 - Efficient computation - just multiply diagonals
- Zero / Ones Matrices
 - Initialization in algorithms

4. Linear Systems

- Solving Linear Equations: $Ax = b$
 - Find x given A and b
 - Regression: Find coefficients that predict outcome
 - Example:
Expression = $\beta_0 + \beta_1 \cdot \text{Genotype} + \beta_2 \cdot \text{Age} + \text{error}$

- Key Concepts

- Invertibility: Does A^{-1} exist?
- Rank: How many independent rows/columns?
- Full-rank vs. rank-deficient matrices

$$A\vec{x} = \vec{b} \text{ where } A = \begin{bmatrix} 1 & 0 & -2 \\ 2 & 1 & -3 \\ 0 & -2 & 1 \\ 4 & 1 & -2 \end{bmatrix} \text{ and } \vec{b} = \begin{bmatrix} 6 \\ 9 \\ 0 \\ 11 \end{bmatrix}$$

- The Omics Problem

- 100 genes, 50 samples $\rightarrow$ more variables than observations
- Matrix is rank-deficient $\rightarrow$ no unique solution!

$$\begin{aligned} 1 \cdot x_1 + 0 \cdot x_2 - 2 \cdot x_3 &= 6 \\ 2 \cdot x_1 + 1 \cdot x_2 - 3 \cdot x_3 &= 9 \\ 0 \cdot x_1 - 2 \cdot x_2 + 1 \cdot x_3 &= 0 \\ 4 \cdot x_1 + 1 \cdot x_2 - 2 \cdot x_3 &= 11 \end{aligned}$$

5. Vector Spaces and Subspaces

- Column Space and Row Space
 - Column space: All possible linear combinations of columns
 - Row space: All possible linear combinations of rows
 - Your data lives in a subspace of the full space
 - Latent space
- Linear Independence vs. Dependence
 - Independent: No gene is a linear combination of others
 - Dependent: Some genes are perfectly correlated
 - Co-expressed genes form dependent sets
- Basis and Dimension
 - Basis: Minimal set of vectors that spans the space
 - Dimension: Number of basis vectors needed
 - Every vector in the space can be expressed as a linear combination of basis vectors.
 - PCA finds an optimal basis for your data (Each subsequent component (maximizes remaining variance, orthogonal to all previous components)

6. Matrix Factorization/Decomposition

- The Central Idea
 - Break complex matrix into simpler pieces: $X = A \cdot B$
 - Each piece captures different aspects of the data
- Eigenvalues and Eigenvectors
 - $Av = \lambda v$ - vector v unchanged in direction by A
 - Eigenvalues = variance explained along each eigenvector
 - Core of PCA and spectral methods
- Singular Value Decomposition (SVD)
 - $X = U \cdot \Sigma \cdot V^T$
 - U : Sample patterns, V : Gene patterns, Σ : Importance
 - Works on ANY matrix (even non-square)
 - Foundation for PCA, MOFA2, Spectral Biclustering, and more

7. Distance and Similarity Measures

- Z-values: how many standard deviations a value is from the zero mean
- Euclidean: $\sqrt{\text{sum of squared differences}}$
- Manhattan: sum of absolute values - robust to outliers
- Cosine similarity: dot product / (norm1 $\times$ norm2)
 - Measures angle between vectors, ignores magnitude

$$\cos(\tilde{x}, \tilde{y}) = \frac{(\tilde{x})^\top (\tilde{y})}{\|\tilde{x}\| \|\tilde{y}\|}$$

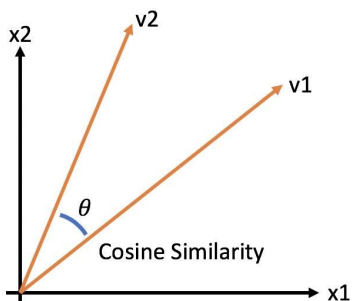
- Correlation: Cosine similarity on centered (subtracting the mean vector) data
- If we center and scale both vectors:

$$\cos(z_x, z_y) = \text{corr}(x, y)$$

$$\|v\| = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}$$

$$v = (3, 4)$$

$$\|v\| = \sqrt{3^2 + 4^2} = 5$$



8. Projections

- Orthogonal Projections
 - Project high-dimensional data onto a subspace
 - Like casting a shadow onto a plane
 - Central to PCA, PLS, and factor analysis
- Projection Matrix
 - $P = X(X^T X)^{-1} X^T$
 - Projects any vector onto the column space of X
 - $P^2 = P$ (projecting twice = projecting once)
- Why Projections Matter
 - PCA: Project onto directions of maximum variance
 - Regression: Project outcome onto predictor space
 - Residuals: What's left after projection

Let

$$X = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

This matrix spans the 1D subspace: the line $y = x$ in $\mathbb{R}^2$.

Step 1: Compute the projection matrix

$$X^T X = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 2$$

$$(X^T X)^{-1} = \frac{1}{2}$$

$$P = X(X^T X)^{-1} X^T = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \frac{1}{2} \begin{bmatrix} 1 & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Step 2: Project a vector

Take

$$y = \begin{bmatrix} 2 \\ 0 \end{bmatrix}$$

$$\hat{y} = Py = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

This is the closest point on the line $y=x$ in squared Euclidean distance

9. Covariance and Correlation

- Covariance Matrix
 - How two variables vary together relative to their means
 - $\text{Cov}(X) = (1/(n-1)) X^T X$ (after centering X)
 - Diagonal: Variance of each gene
 - Off-diagonal: Covariance between gene pairs

$$X = \begin{bmatrix} -1 & -2 \\ 0 & 0 \\ 1 & 2 \end{bmatrix}$$

- Correlation Matrix
 - Standardized covariance: $\text{Corr} = \text{Cov} / (\text{SD1} \times \text{SD2})$
 - Values always between -1 and +1
 - Interpretation: +1 = perfectly correlated, -1 = anti-correlated

$$\text{Cov}(X) = \frac{1}{n-1} X^T X = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

- Applications
 - WGCNA: Builds networks from correlation matrix
 - PCA: Eigendecomposition of covariance matrix
 - Clustering: Groups genes by correlation patterns

$$\text{Cov}(X) = \begin{bmatrix} \text{Var}(x) & \text{Cov}(x, y) \\ \text{Cov}(y, x) & \text{Var}(y) \end{bmatrix}$$

10. Sparse and High-Dimensional Matrices

- Understanding Sparsity
 - Most entries are zero or near-zero
 - Example: Single-cell RNA-seq - 80-90% zeros
 - Not all genes are expressed in all cells
- Implications for Analysis
 - Storage: Sparse matrix formats save memory
 - Speed: Sparse algorithms skip zero computations
 - Statistics: Standard methods may not apply
- The Curse of Dimensionality
 - 20,000 genes, 300 samples: More variables than observations
 - Standard regression can't handle this
 - Need regularization or dimensionality reduction

11. Graphs

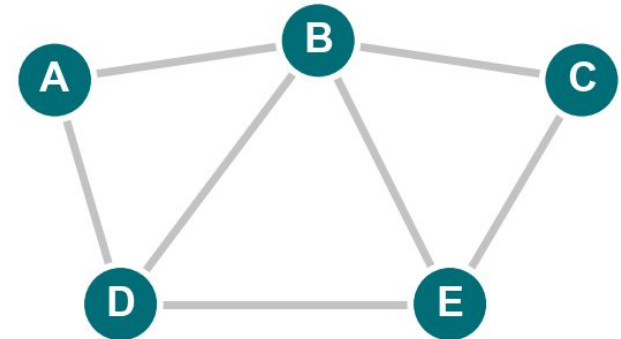
$G = (V, E)$

$V = \{v_1, v_2, \dots, v_n\}$ - set of vertices
(nodes)

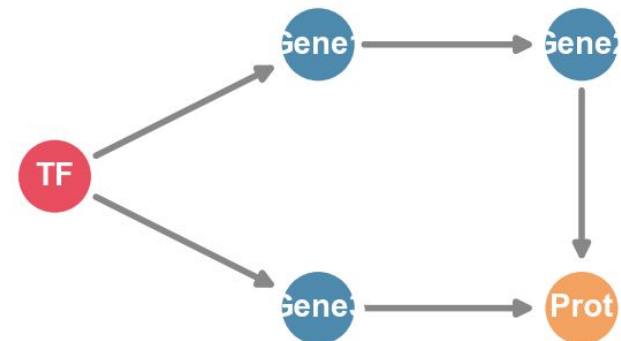
$E = \{(v_i, v_j)\}$ - set of edges (links
between nodes)

- ▶ Represent interaction in biological systems
- ▶ Genes, proteins, metabolites form complex networks
- ▶ Graph theory provides a rigorous mathematical language
- ▶ Enables computational analysis of biological topology
- ▶ Unifies diverse omics layers into a single framework

Undirected Graph



Directed Graph (Digraph)



11. Graphs

Degree $\deg(v)$

Number of edges incident to vertex v .

In a directed graph: in-degree (edges in)

and out-degree (edges out)

Hub genes in co-expression networks have high degree (connectivity).

Adjacency Matrix A

An $n \times n$ matrix where $A[i][j] = 1$ if edge (v_i, v_j) exists, 0 otherwise. Symmetric for undirected graphs.

WGCNA builds a weighted adjacency matrix from expression correlations.

Path & Shortest Path

A sequence of vertices connected by edges. Shortest path = minimum edges (or total weight) between two nodes.

Signaling cascades are paths in protein interaction networks.

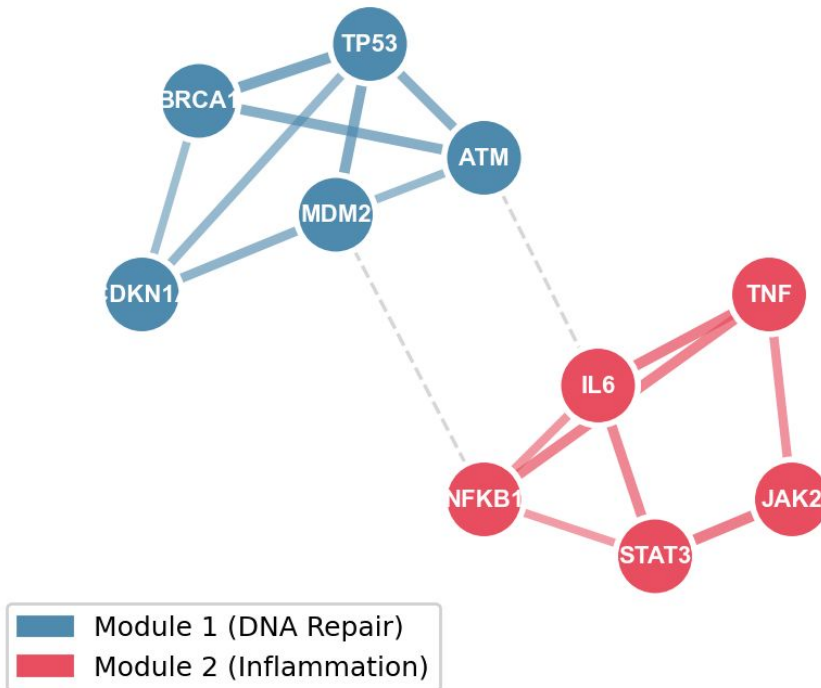
Connected Component

A maximal subgraph where every pair of vertices is reachable from every other via some path.

Gene modules / clusters are often identified as connected components.

11. Graphs

Weighted Co-expression Network (WGCNA-style)



Graph Types in Biological Networks

Weighted Graph

$G = (V, E, w)$ where
 $w: E \rightarrow \mathbb{R}$

Edge weights
represent correlation
strength (e.g.
WGCNA adjacency)

Directed Acyclic Graph (DAG)

Directed graph with no
cycles;
topological ordering
exists

Gene Ontology
hierarchy,
causal Bayesian
networks

Bipartite Graph

$V = U \cup W$, $E \subseteq U \times W$;
no edges within U or
 W

eQTL SNP-gene
mappings,
(edges only between
groups)

Hypergraph

Edges connect any
number of
vertices (not just pairs)

Protein complexes,
metabolic
reactions with
multiple substrates

Multiplex / Multi-layer Graph

Same vertex set,
multiple edge
sets (one per omics
layer)

Integrated
multiomics: same
genes,
different relation
types per layer

Linear Algebra Summary

- Core Structures
 - Vectors: Individual samples or features
 - Matrices: Complete datasets (samples $\times$ features)
- Essential Operations
 - Transpose, multiplication, decomposition
 - These power all our analysis methods
- Key Decompositions
 - SVD/PCA: Find structure in high-dimensional data
 - Used in MOFA2, WGCNA, and ML methods
- Why This Matters
 - Understanding the math helps you debug analyses
 - Knowing limitations prevents misuse
 - We'll apply all of this in our practicals

Example: Instability of Fold Change Under Different Preprocessing

Group 1 (Gene A = [61, 24, 81, 70, 30, 92, 96], n=7): mean = 64.86

Group 2 (Gene A = [84, 84, 97, 33, 12], n=5): mean = 61.80

1. Raw fold change: $FC = 61.80 / 64.86 \approx 0.96$ (slightly downregulated)
2. After normalization (by total counts):

Group 1 mean = 0.1429

Group 2 mean = 0.2000

$FC \approx 1.40$ (upregulated)

3. After $\log_2(x + 1)$ transformation:

Group 1 mean = 6.32

Group 2 mean = 6.08

$\Delta\log_2 \approx -0.24$ (moderately downregulated)

Conclusion: Fold change is unstable without clearly defined normalization and transformation.
Use DESeq2: shrinkage $\log_2FC$, edgeR: GLM-based $\logFC$, limma-voom: moderated $\logFC$

Multionics data preparation workflow

Data preparation

- WGS
 - FASTQ → alignment (BWA-MEM2)
 - duplicate marking
 - BQSR
 - Haplotype Caller & Joint variant calling
- RNA-seq
 - FASTQ → alignment (STAR 2-pass)
 - gene & splice quantification
- Inputs:
 - Multi-sample VCF file from joint calling
 - Raw gene expression (bulk RNA) - with matching sample IDs with DNA
 - Covariates CSV (also matching samples)
 - Gene annotation file (GTF) for defining cis-windows

Sample validation

- Extract sample IDs from VCF, expression matrix, and covariates file
- Identify intersection of samples present in all three files
- Report excluded samples and warn if >10% are missing from any file
- Subset all files to common samples only
- Validate genome build consistency between VCF and GTF (GRCh37 vs GRCh38)
- Harmonize chromosome naming convention (chr1 vs 1) between VCF and GTF

DNA processing, constraints and filtering

- Biallelic variants only: required for additive linear QTL models
- Restrict to autosomes (chr1-22): removes sex chromosomes and mitochondrial variants to avoid sex-specific eQTL confounding
- Standardize variant IDs to chr:pos:ref:alt format: ensures unique identifiers when rsIDs are missing or duplicated
- MAF filtering (≥ 0.05 default, cmd line parameter; 0.01 is common for larger samples): ensures statistical power and model stability
- Call rate $\geq 95\%$ (default, cmd line parameter): avoids bias from missing genotypes
- Sample QC: heterozygosity check and relatedness filtering (remove duplicates/relatives if needed)

RNA processing, constraints and filtering

- Match gene IDs between expression matrix and GTF (handle Ensembl version suffixes like ENSG00000123456.1 → ENSG00000123456)
- Filter to protein-coding genes only (optional, cmd line parameter): focuses analysis on well-annotated genes
- Filter low-expression genes (CPM > 1 in at least 20% of samples default, cmd line parameter)
- Detect expression outliers via PCA (samples >3 SD from mean on PC1-3) and report to user in QC summary
- Log2(CPM+1) transformation: stabilizes variance across expression range
- TMM normalization: adjusts for composition bias between samples
- Fit PEER to regress out hidden confounders with K factors
- Extract PEER factors (samples × K) and add to covariates
- Apply inverse normal transformation to residuals: enforces Gaussian distribution for QTL mapping

Initial Scan with Maximal Filtering (Recommended)

- Use strict MAF (≥ 0.05) to maximize power per test
- Restrict to protein-coding genes to reduce dimensionality
- Run fewer PEER factors to detect dominant signals
- Purpose: validate pipeline, detect batch effects, estimate signal-to-noise

Understanding Multi-Omics Data

Multi-Omics Data & Course Dataset

- What is Multi-Omics?
 - Genomics: DNA sequence, genetic variants (SNPs)
 - Transcriptomics: RNA expression levels (which genes are active)
 - Integration: Link variants → expression → phenotype
- Our ASD Dataset: 300 samples (200 cases, 100 controls)
 - Expression: 300 × 100 genes (ASD_expression.csv)
 - Genotypes: 300 × 50 SNPs (ASD_genotypes.csv)
 - Covariates: Age, sex, IQ, ADOS scores (ASD_covariates.csv)
- Key Autism Genes: SHANK3, CHD8, NRXN1, MECP2...
 - Synaptic function, chromatin remodeling, neuronal signaling

Data Loading and Integration

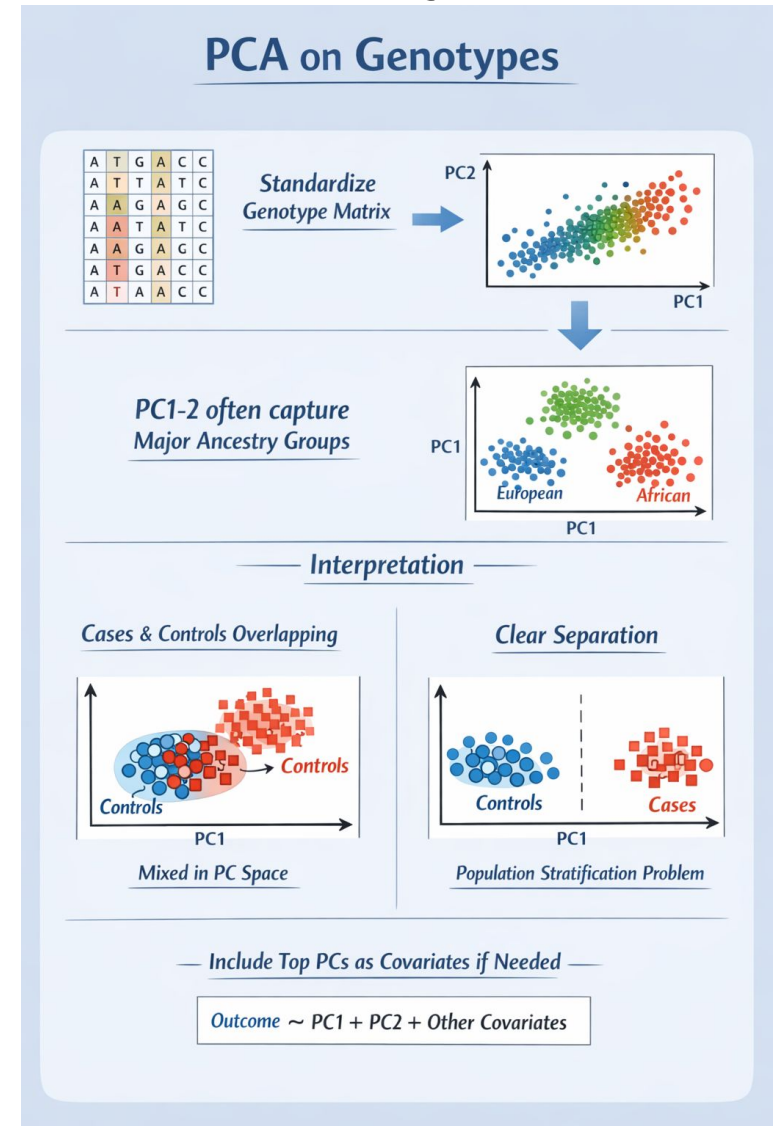
- Loading Multi-Layer Data
 - Covariates: `pd.read_csv('ASD_covariates.csv')`
 - Genotypes: `pd.read_csv('ASD_genotypes.csv')`
 - Expression: `pd.read_csv('ASD_expression.csv')`
- Sample Alignment (Critical!)
 - Set sample ID as index in all DataFrames
 - Verify: `covariates.index.equals(genotypes.index)`
 - Verify: `genotypes.index.equals(expression.index)`
- Shape Verification
 - Expression: (300, 100) - 300 samples, 100 genes
 - Genotypes: (300, 50) - 300 samples, 50 SNPs
 - Covariates: (300, 29) - 300 samples, 29 variables

Check Data

- Diploid Genotype Representation
 - 0 = Homozygous reference (REF/REF) - common genotype
 - 1 = Heterozygous (REF/ALT) - one copy of variant
 - 2 = Homozygous alternate (ALT/ALT) - two copies
- Quality Checks
 - Values should only be 0, 1, or 2
 - Check: `genotypes.isin([0,1,2]).all().all()`
 - No missing values (or handle explicitly)
 - Outlier detection (PCA, clustering)
 - Unexpected sex (check sex-specific genes)
 - Sample swaps (common in large studies!)
- Allele Frequency Metrics
 - Allele frequency = $\text{mean}(\text{dosage}) / 2$
 - MAF = $\min(\text{freq}, 1 - \text{freq})$ - always ≤ 0.5
 - Filter SNPs with MAF < 0.01 (too rare)

Population Structure Analysis

- Why Check Population Structure?
 - Genetic variants differ by ancestry
 - Expression levels differ by ancestry
 - If cases/controls have different ancestry → spurious results
- PCA on Genotypes
 - Standardize genotype matrix
 - Compute principal components
 - PC1-2 often capture major ancestry groups
- Interpretation
 - Cases and controls should overlap in PC space
 - Clear separation = population stratification problem
 - Include top PCs as covariates if needed



Covariate Balance Testing

- Age Distribution
 - T-test comparing ASD vs Control
 - Report: Mean $\pm$ SD for each group, p-value
 - Significant difference? Adjust for it!
- Sex Distribution
 - Chi-square test of independence
 - Expected: ~4:1 male bias in ASD
 - Report: % male by group, p-value
- IQ Distribution
 - T-test comparing groups
 - ASD often has lower mean IQ (heterogeneous)
 - Important confounder for expression analysis

Visualizing Sample Characteristics

- 6-Panel Comparison Plot
 - Panel 1: Age distribution (histograms by group)
 - Panel 2: IQ distribution
 - Panel 3: Sex distribution (bar chart)
 - Panel 4: ADOS scores (ASD only)
 - Panel 5: Social impairment scores
 - Panel 6: Repetitive behavior scores
- Additional QC Plots
 - Expression: Overall distribution + per-gene boxplots
 - Genotypes: MAF distribution + example genotypes
 - PCA: Scatter plot (PC1 vs PC2) + scree plot

Why Multi-Omics Integration?

Why Multi-Omics Integration?

- Single-Omics Limitations
 - Genomics: 'SNP rs123 is associated with ASD' - but how?
 - Transcriptomics: 'Gene X is differentially expressed' - but why?
- Integration Answers 'How' and 'Why'
 - rs123 increases Gene X expression → ASD risk
 - Now we have a mechanism!
 - Can target Gene X for therapy
- Course Methods for Integration
 - eQTL: Direct SNP → Expression links
 - WGCNA: Find co-regulated gene modules
 - MOFA2: Discover shared factors across omics
 - ML: Predict outcomes using all data layers

Multi-Omics Data Integration Framework

- Data Layers in Our Analysis
 - Layer 1: Genotypes (300×50) - inherited, static
 - Layer 2: Expression (300×100) - dynamic, tissue-specific
 - Layer 3: Covariates (300×29) - clinical context
- Integration Approaches
 - Early integration: Concatenate features, then analyze
 - Late integration: Analyze separately, combine results
 - Intermediate: MOFA2 - find shared latent factors
- The Central Equation
 - Phenotype = $f(\text{Genotype}, \text{Expression}, \text{Covariates})$
 - Our goal: Learn f and identify key features

Good practices

- Golden rule of bioinformatics: **Never ever trust your tools (or data)**
- Adapt robust and reproducible practices
 - Document each step
 - Write down key facts
 - Conclude the work done to easy understand it again in few months (make figures, reports)
 - Automate manual tasks with script
 - Always ask yourself: How much time would it take me to do it one more time?
 - Release or publish your code and tools
- Write code for humans, Write (meta)data for computers
 - Make assertions (be loud)
 - Let the code test the code
 - Use existing libraries whenever possible
 - Frequently used scripts -> tools
 - Use code versioning (git)

Summary: Lesson 1 Complete!

- What We Covered Today
 - Linear algebra foundations for omics analysis
 - Multi-omics data types (genomics, transcriptomics)
 - ASD dataset overview and structure
 - Data loading, QC, and validation
- Key Takeaways
 - Multi-omics integration reveals mechanisms
 - QC is essential - garbage in, garbage out
 - Sample alignment is critical for integration
 - Understand your data before analyzing it
- You Can Now
 - Load and verify multi-omics datasets
 - Perform basic QC checks
 - Interpret linear algebra in biological context

Questions?

- Review Materials
 - Notebook: notebooks/1_intro.ipynb
 - Data: data/ASD_dataset/ directory
 - Slides: Available on course Github
- Next Session Preview
 - Quantitative Trait Loci (QTL) Analysis
 - Expression QTL (eQTL) Mapping
 - Statistical testing and multiple comparisons